Data Visualization Project - R

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2021-12-14
Last compilation on 2022-01-11

1 Installing Packages required for the task and creating working directory

```
BiocManager::install("ggplot2")
BiocManager::install("ggrepel")
dir.create("~/Desktop/Data_Viz_Project", recursive = TRUE)
setwd("~/Desktop/Data_Viz_Project")
```

2 Libraries

```
library(ggplot2)
library(ggrepel)
library(dplyr)
library(tidyverse)
library(ggpubr)
library(pheatmap)
library(RColorBrewer)
```

3 TCGA expression data

3.1 Load and prepare data

```
download.file("https://raw.githubusercontent.com/lauanna/Bioinformatics-Tools/refs/heads/main/Data_Viz_Project/TCGA_expression_matrix.txt", "~/Desktop/Data_Viz_Project/TCGA_expression_matrix.txt")
download.file("https://raw.githubusercontent.com/lauanna/Bioinformatics-Tools/refs/heads/main/Data_Viz_Project/gene_lengths.txt", "~/Desktop/Data_Viz_Project/gene_lengths.txt")

TCGA_expr <- read.table("~/Desktop/Data_Viz_Project/TCGA_expression_matrix.txt", header = TRUE, sep = "\t")
gene_lengths <- read.table("~/Desktop/Data_Viz_Project/gene_lengths.txt", header = TRUE, sep = "\t")

head(TCGA_expr, n = 3L)
head(gene_lengths, n = 3L)

##   gene_id Control_1 Control_2 Control_3 Hypoxia_1 Hypoxia_2 Hypoxia_3
## 1  ENSG00     1.000     1.000     1.000     1.000     1.000     1.000
## 2  ENSG00     1.000     1.000     1.000     1.000     1.000     1.000
## 3  ENSG00     1.000     1.000     1.000     1.000     1.000     1.000
##   gene_id gene_length
## 1  ENSG00        2473
## 2  ENSG00        1867
## 3  ENSG00        1589

dim(TCGA_expr)
dim(gene_lengths)

## [1] 19962     7
## [1] 19962     2
```

3.2 Plotting

3.2.1 Expression distribution (Liver)

```
ggplot(data = TCGA_expr, aligned = TRUE)
  + geom_histogram(aes(x = Control_1), fill = "#ffb3ff", color = "black")
  + theme(panel.background = element_rgb(0.9, 0.9, 0.9))
  + ggtitle("Expression Gene 1")
  + xlab("Read counts")
  + ylab("Count")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank())
  + theme(axis.ticks.y = element_blank())
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank())
  + theme(panel.grid.major.y = element_blank()) ## ggsave("./Expression_Gene_1.pdf", dpi = 300, width = 5, height = 5)
```

We wanted to compare the expression of a gene in order to determine the significance of each technical replicate in our data. Assuming the Hypothesis that the expression of a gene should be similar in all technical replicates, we decided to plot the expression of one of the genes.

We chose the histogram because it allows us to see the distribution of the data and it is a simple way to see if there is any significant difference between the expression of the gene in the different technical replicates.

We chose to use ggplot because it allows us to create beautiful and aesthetic plots.

Finally we added a title to the plot, changed the color of the bars and made the axis more clear.

```
ggplot(data = TCGA_expr, aligned = TRUE)
  + geom_histogram(aes(x = Control_1), fill = "#ffb3ff", color = "black")
  + theme(panel.background = element_rgb(0.9, 0.9, 0.9))
  + ggtitle("Expression Gene 1")
  + xlab("Read counts")
  + ylab("Count")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Expression_Gene_1.pdf", dpi = 300, width = 5, height = 5)
```

```
ggplot(data = TCGA_expr, aligned = TRUE)
  + geom_histogram(aligned = TRUE, aes(x = Control_1), fill = "#ffb3ff", color = "black")
  + theme(panel.background = element_rgb(0.9, 0.9, 0.9))
  + ggtitle("Expression Gene 1")
  + xlab("Read counts")
  + ylab("Count")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Expression_Gene_1.pdf", dpi = 300, width = 5, height = 5)
```

3.2.2 Gene expression correlation

```
ggplot(data = TCGA_expr, aligned = TRUE, mapping = aes(x = Control_1, y = Hypoxia_1))
  + geom_point(aes(color = "#ffb3ff"))
  + geom_smooth(method = "lm", se = FALSE, aes(color = "black"))
  + stat_cor(method = "pearson", aes(color = "black"))
  + theme(panel.background = element_rgb(0.9, 0.9, 0.9))
  + ggtitle("Relationship Gene 1 and Gene 12")
  + xlab("Gene 1")
  + ylab("Gene 12")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Relationship_Gene_1_and_Gene_12.pdf", dpi = 300, width = 5, height = 5)
```

In order to find out if there is any correlation between the expression of two genes, we decided to plot a scatter plot. We chose to use a scatter plot because it allows us to see if there is any correlation between the expression of two genes.

We added a regression line to the plot in order to make it easier to see the correlation between gene 1 and gene 12. We also added the Pearson correlation coefficient to the plot in order to make it easier to see the correlation between the two genes.

Finally we changed the plot aesthetics to make it more aesthetic and easier to read. We changed the color of the bars, added a title to the plot, changed the axis titles and changed the axis ticks to make it more aesthetic.

3.2.3 Chi-square proportions

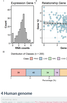
```
chi_data <- c(1, 2, 3, 4)
chi_data <- matrix(chi_data, nrow = 2, byrow = TRUE, dimnames = c(c("Gene 1", "Gene 2"), c("Gene 3", "Gene 4")))
chi_data
chi.test(chi_data, correction = FALSE)
chi.test(chi_data, correction = TRUE)

##     Gene 3 Gene 4
## Gene 1      1      3
## Gene 2      2      4
##
## Pearson's Chi-squared test
## data:  chi_data
## X-squared = 0.2, df = 1, p-value = 0.6547
##
## Pearson's Chi-squared test with Yates' continuity correction
## data:  chi_data
## X-squared = 0, df = 1, p-value = 1
```

We used chi-square to see if there is any significant difference between the proportions of two genes. We chose to use chi-square because it allows us to see if there is any significant difference between the proportions of two genes.

We added a title to the plot, changed the axis titles and changed the axis ticks to make it more aesthetic and easier to read.

3.2.4 Combine the plots

```
ggarrange(Expression_Gene_1, Relationship_Gene_1_and_Gene_12, Distribution_of_Classes_1-100, ncol = 3, nrow = 1, labels = c("A", "B", "C"))
```



4 Human genome

```
BiocManager::install("BSgenome.Hsapiens.UCSC.hg19")
BiocManager::install("TxDb.Hsapiens.UCSC.hg19.knownGene")
BiocManager::install("org.Hs.eg.db")
BiocManager::install("ChIPseeker")

##  ggplot2 3.4.0   2022-11-28 CRAN (R 4.2.2)
##  ggrepel 0.9.2   2022-11-06 CRAN (R 4.2.2)
##  ggsci   2.9     2018-05-28 CRAN (R 4.2.2)
##  ggsignif 0.6.4  2022-10-13 CRAN (R 4.2.2)
##  gtable  0.3.0   2019-03-25 CRAN (R 4.2.2)
##  gtools  3.9.4   2022-11-27 CRAN (R 4.2.2)
##  hms     1.1.2   2022-08-19 CRAN (R 4.2.2)
##  htmlwidgets 1.5.4 2021-09-08 CRAN (R 4.2.2)
##  jsonlite 1.8.4  2022-12-06 CRAN (R 4.2.2)
##  KernSmooth 2.23-20 2021-05-03 CRAN (R 4.2.2)
##  lattice 0.20-45 2021-09-22 CRAN (R 4.2.2)
##  lifecycle 1.0.3 2022-10-07 CRAN (R 4.2.2)
##  magrittr 2.0.3  2022-03-30 CRAN (R 4.2.2)
##  MASS    7.3-58.1 2022-08-03 CRAN (R 4.2.2)
##  Matrix  1.5-3   2022-11-07 CRAN (R 4.2.2)
##  mgcv    1.8-41  2022-10-21 CRAN (R 4.2.2)
##  munsell 0.5.0   2018-06-12 CRAN (R 4.2.2)
##  nlme    3.1-162 2022-10-07 CRAN (R 4.2.2)
##  pillar  1.9.0   2023-03-22 CRAN (R 4.2.2)
##  pkgconfig 2.0.3 2019-09-22 CRAN (R 4.2.2)
##  plyr    1.8.8   2022-11-11 CRAN (R 4.2.2)
##  polyclip 1.10-4 2022-10-20 CRAN (R 4.2.2)
##  prettyunits 1.1.1 2020-01-24 CRAN (R 4.2.2)
##  purrr   1.0.1   2023-01-10 CRAN (R 4.2.2)
##  R6      2.5.1   2021-08-19 CRAN (R 4.2.2)
##  RBGL    1.74.0  2022-11-01 Bioc (R 4.2.2)
##  RCurlCurl 1.95-4.12 2022-09-28 CRAN (R 4.2.2)
##  rlang   1.1.0   2023-03-14 CRAN (R 4.2.2)
##  rstatix 0.7.2   2023-02-01 CRAN (R 4.2.2)
##  rtracklayer 1.58.0 2022-11-01 Bioc (R 4.2.2)
##  rvest   1.0.3   2022-08-19 CRAN (R 4.2.2)
##  scales  1.2.1   2022-08-20 CRAN (R 4.2.2)
##  scatterplot3d 0.3-41 2023-04-03 CRAN (R 4.2.2)
##  scatterpie 0.1.8 2023-03-02 CRAN (R 4.2.2)
##  shadowtext 0.1.2 2022-04-22 CRAN (R 4.2.2)
##  shape   1.4.6   2021-05-19 CRAN (R 4.2.2)
##  stringi 1.7.12  2023-01-11 CRAN (R 4.2.2)
##  stringr 1.5.0   2022-12-02 CRAN (R 4.2.2)
##  tibble  3.2.1   2023-03-20 CRAN (R 4.2.2)
##  tidy.gff2 1.0.2 2022-11-01 Bioc (R 4.2.2)
##  tidyr   1.3.0   2023-01-24 CRAN (R 4.2.2)
##  tidyselect 1.2.0 2022-10-10 CRAN (R 4.2.2)
##  tidytree 0.4.2  2022-12-18 CRAN (R 4.2.2)
##  tzdb    0.3.0   2022-11-16 CRAN (R 4.2.2)
##  utf8    1.2.3   2023-01-31 CRAN (R 4.2.2)
##  vctrs   0.6.1   2023-03-22 CRAN (R 4.2.2)
##  vroom   1.6.0   2023-01-27 CRAN (R 4.2.2)
##  withr   2.5.0   2022-03-03 CRAN (R 4.2.2)
##  XML     3.99-0.13 2022-11-22 CRAN (R 4.2.2)
##  zip     2.2.2   2022-11-11 CRAN (R 4.2.2)
##  zoo     1.8-12  2023-04-13 CRAN (R 4.2.2)
##
## [1] /home/rstudio/tmp/Rtmp8Ldqqq/Rlibs
## [2] /usr/local/lib/R/site-library
## [3] /usr/lib/R/library
##
## ------------------------------------------------------------------------
## Warning messages:
## 1: In .Deprecated("BiocManager::install(ask=FALSE)") :
##  'BiocManager::install(ask=FALSE)' is deprecated.
## Use 'BiocManager::install(ask=TRUE)' instead.
## 2: In .Deprecated("BiocManager::install(ask=FALSE)") :
##  'BiocManager::install(ask=FALSE)' is deprecated.
## Use 'BiocManager::install(ask=TRUE)' instead.
## 3: In .Deprecated("BiocManager::install(ask=FALSE)") :
##  'BiocManager::install(ask=FALSE)' is deprecated.
## Use 'BiocManager::install(ask=TRUE)' instead.
## 4: In .Deprecated("BiocManager::install(ask=FALSE)") :
##  'BiocManager::install(ask=FALSE)' is deprecated.
## Use 'BiocManager::install(ask=TRUE)' instead.
```

4.2 Plotting

4.2.1 Gene count

```
ggplot(data = Mouse_Human_Genes_Count, aligned = TRUE)
  + geom_bar(aes(x = Species, fill = Species))
  + ggtitle("Total gene number")
  + xlab("Species")
  + ylab("Count")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Total_gene_number.pdf", dpi = 300, width = 5, height = 5)
```

4.2.2 Length distribution

```
ggplot(data = Mouse_Human_Genes_Length, aligned = TRUE)
  + geom_density(aes(x = Gene_Length, fill = Species))
  + ggtitle("Length distribution of gene classes")
  + xlab("Gene Length (bp)")
  + ylab("Density")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Length_distribution_of_gene_classes.pdf", dpi = 300, width = 5, height = 5)
```

In this plot we wanted to see the distribution of the length of the genes. We chose a density plot because it allows us to see the distribution of the data and it is a simple way to see if there is any significant difference between the distribution of the length of the genes in the different species.

We added a title to the plot, changed the axis titles and changed the axis ticks to make it more aesthetic and easier to read.

```
ggplot(data = Mouse_Human_Genes_Length, aligned = TRUE)
  + geom_density(aes(x = Gene_Length, fill = Species))
  + ggtitle("Length distribution of gene classes")
  + xlab("Gene Length (bp)")
  + ylab("Density")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Length_distribution_of_gene_classes.pdf", dpi = 300, width = 5, height = 5)
```

4.2.3 Combine figures

```
ggarrange(Total_gene_number, Length_distribution_of_gene_classes, ncol = 2, nrow = 1, labels = c("A", "B"))
```

5 Differential gene Expression

5.1 Load and prepare data

```
download.file("https://raw.githubusercontent.com/lauanna/Bioinformatics-Tools/refs/heads/main/Data_Viz_Project/Differential_Expression_Results.txt", "~/Desktop/Data_Viz_Project/Differential_Expression_Results.txt")

Diff_Expr <- read.table("~/Desktop/Data_Viz_Project/Differential_Expression_Results.txt", header = TRUE, sep = "\t")

head(Diff_Expr, n = 3L)

##   gene_id baseMean log2FoldChange lfcSE stat pvalue padj
## 1  ENSG00     0.000             NA    NA   NA     NA   NA
## 2  ENSG00     0.000             NA    NA   NA     NA   NA
## 3  ENSG00     0.000             NA    NA   NA     NA   NA

dim(Diff_Expr)

## [1] 19962     7
```

5.2 Plotting

5.2.1 Volcano plot

```
ggplot(data = Diff_Expr, aligned = TRUE)
  + geom_point(aes(x = log2FoldChange, y = -log10(pvalue)))
  + ggtitle("Diff. expressed genes (Cancer vs. Normal)")
  + xlab("Fold change (log2)")
  + ylab("Adjusted p-value (-log10)")
  + theme(plot.title = element_text(hjust = 0.5))
  + theme(axis.text.x = element_text(angle = 0, hjust = 0.5))
  + theme(axis.text.y = element_text(angle = 0, hjust = 1))
  + theme(axis.ticks.x = element_blank)
  + theme(axis.ticks.y = element_blank)
  + theme(axis.line.x = element_line(color = "black"))
  + theme(axis.line.y = element_line(color = "black"))
  + theme(axis.title.x = element_text(color = "black"))
  + theme(axis.title.y = element_text(color = "black"))
  + theme(axis.text.x = element_text(color = "black"))
  + theme(axis.text.y = element_text(color = "black"))
  + theme(panel.grid.major.x = element_blank)
  + theme(panel.grid.major.y = element_blank) ## ggsave("./Diff._expressed_genes_(Cancer_vs._Normal).pdf", dpi = 300, width = 5, height = 5)
```

5.2.2 Heatmaps

```
Diff_Expr_Top_Genes <- head(Diff_Expr[order(Diff_Expr$padj), ], n = 50L)
Diff_Expr_Top_Genes_Matrix <- as.matrix(Diff_Expr_Top_Genes[, c("Control_1", "Control_2", "Control_3", "Hypoxia_1", "Hypoxia_2", "Hypoxia_3")])
rownames(Diff_Expr_Top_Genes_Matrix) <- Diff_Expr_Top_Genes$gene_id
Diff_Expr_Top_Genes_Matrix <- t(scale(t(Diff_Expr_Top_Genes_Matrix)))
pheatmap(Diff_Expr_Top_Genes_Matrix, cluster_rows = TRUE, cluster_cols = TRUE, show_rownames = TRUE, show_colnames = TRUE, scale = "none", clustering_distance_rows = "euclidean", clustering_distance_cols = "euclidean", clustering_method = "complete", cutree_rows = NA, cutree_cols = NA, legend = TRUE, annotation_row = NA, annotation_col = NA, annotation_legend = TRUE, annotation_names_row = TRUE, annotation_names_col = TRUE, drawing_width = NA, drawing_height = NA, gaps_row = NULL, gaps_col = NULL, labels_row = NULL, labels_col = NULL, filename = NA, silent = FALSE, fontsize = 10, fontsize_row = 10, fontsize_col = 10, display_numbers = FALSE, number_format = "%.2f", fontsize_number = 8, cellwidth = NA, cellheight = NA, angle_col = 270, treeheight_row = 50, treeheight_col = 50, breaks = NA, color = colorRampPalette(c("#ffb3ff", "#ffffff", "#ffff00"))(100), cellnote = NA, cellnote_color = "black", cellnote_fontsize = NA, tooltip = NA, clustering_callback = function(hclust, mat) { hclust }, gaps_from_clustering = NULL, gaps_from_annotations = NULL, annotation_colors = NA, annotation_legend_side = "bottom", annotation_row_side = "left", annotation_col_side = "top", annotation_row_order = NULL, annotation_col_order = NULL, annotation_row_gap = 3, annotation_col_gap = 3, annotation_row_gap_size = 3, annotation_col_gap_size = 3, annotation_row_gap_color = "black", annotation_col_gap_color = "black", annotation_row_gap_type = "solid", annotation_col_gap_type = "solid", annotation_row_gap_width = 1, annotation_col_gap_width = 1, annotation_row_gap_alpha = 1, annotation_col_gap_alpha = 1, annotation_row_gap_lty = 1, annotation_col_gap_lty = 1, annotation_row_gap_lwd = 1, annotation_col_gap_lwd = 1, annotation_row_gap_col = "black", annotation_col_gap_col = "black", annotation_row_gap_pch = 1, annotation_col_gap_pch = 1, annotation_row_gap_cex = 1, annotation_col_gap_cex = 1, annotation_row_gap_bg = "black", annotation_col_gap_bg = "black", annotation_row_gap_pch_bg = 1, annotation_col_gap_pch_bg = 1, annotation_row_gap_cex_bg = 1, annotation_col_gap_cex_bg = 1, annotation_row_gap_lty_bg = 1, annotation_col_gap_lty_bg = 1, annotation_row_gap_lwd_bg = 1, annotation_col_gap_lwd_bg = 1, annotation_row_gap_col_bg = "black", annotation_col_gap_col
```